TOPIC 16/60

What is Temperature?

> The "creativity" knob for LLMs.
>
> How we control randomness vs logic.

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THE GOAL

When an LLM predicts the next word, it never predicts just *one* word. It predicts a **list of possibilities**, each with a percentage chance.

Prompt: "The sky is..."

Blue (90%)

Clear (7%)

Dark (3%)

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THE CREATIVITY KNOB

🌡️

Temperature

Temperature is a mathematical setting that reshapes these probabilities before the model makes its choice.

> It determines how "adventurous" the model is. Do we play it safe, or do we get wild?

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LOW TEMPERATURE (0.1 - 0.3)

Deterministic

Low temperature makes the model "confident." It aggressively boosts the probability of the most likely word and crushes the others.

Use Cases:

Coding, Data Extraction, Math, Medical diagnosis. You want the most logical, repetitive, correct answer, not the "creative" one.

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HIGH TEMPERATURE (0.7 - 1.2)

Creative

High temperature levels the playing field. It gives the "unlikely" words a higher chance to be picked.

> **Use Cases:** Creative Writing, Brainstorming, Poetry, Storytelling. It adds variety and prevents the model from repeating the same common phrases.

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HOW IT WORKS

Logits

The "Raw" Scores

Before probabilities, the AI outputs "Logits"—a raw score for every word in its dictionary.

Formula: Logit / Temperature

Dividing the logits by a small temp (e.g., 0.1) creates huge gaps between top and bottom scores. Dividing by a high temp (e.g., 1.5) makes them all look equal.

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THE TRADE-OFF

TOO LOW

Repetitive loops.

The model gets "stuck" saying the same phrase over and over because it's too scared to pick anything but the top choice.

TOO HIGH

Nonsense.

The model starts picking completely illogical words, leading to gibberish output.

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VISUALIZATION

Softmax

Temperature is applied right before the Softmax function, which forces all those raw logits to turn into a clean 0% to 100% probability.

```
Low Temp = "Winner takes all"
High Temp = "Everything gets a chance"
```

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TEMPERATURE TIPS

Start at 0.7

This is the sweet spot for general conversation—balanced and natural.

Use 0 for Accuracy

If you are doing data analysis, code, or factual research, set Temperature to 0.

Use > 1.0 for Brainstorming

If you need truly weird, out-of-the-box ideas, push it above 1.

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Temperature

CONCEPT UNLOCKED

- 🌡️ The creativity knob
- 🧊 Low = Deterministic
- 🔥 High = Creative
- 🧮 Adjusts word probabilities

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